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Intermediate Goods and Business Cycles: Implications for Productivity and Welfare

By SUSANTO BASU*

This paper presents an aggregate demand-driven model of business cycles that provides a new explanation for the procyclicality of productivity and simultaneously predicts large welfare losses from monetary nonneutrality. The key features of the model are an input–output production structure, imperfect competition, countercyclical markups, and for some results, state-dependent price rigidity. True technical efficiency is procyclical even though production takes place with constant returns, without technology shocks or technological externalities. The paper has observable implications that distinguish it empirically from related work. These implications are generally supported by data from U.S. manufacturing industries. (*JEL* E32, E23)

This paper studies a business-cycle model with imperfect competition where intermediate goods are used in production. It is an example of a class of models in which markups are countercyclical. One major result is that in this setting, demand-driven output movements cause productivity to be procyclical. The paper studies a number of theoretical and empirical implications of this source of productivity fluctuations. In a subset of models, countercyclical markups result from assuming that there are fixed costs of changing nominal prices. The paper shows that modeling the use of intermediate goods in this type of model greatly expands the extent of price rigidity, leading to larger welfare losses from business cycles.

*Department of Economics, University of Michigan, Ann Arbor, MI 48109. This is a revised version of Chapter 1 of my unpublished 1992 Harvard dissertation. I thank, without implicating, Larry Ball, Bob Barsky, Steve Cecchetti, Brad De Long, John Fernald, John Leahy, Valerie Ramey, David Romer, the members of the Harvard Macro Lunch group, seminar participants at various universities and the NBER, and two anonymous referees, for helpful comments. Special thanks to my advisor, Greg Mankiw, for his unstinting help, advice, and encouragement. Parts of this work were supported by fellowships from the National Science Foundation, the Chiles Foundation, the Social Sciences Research Council, and the NBER, which are gratefully acknowledged.

It is an old idea that an industrialized economy, with its greater interdependence and more roundabout production, is more subject to cyclical output fluctuations. The idea has been present at least since the work of Gardiner C. Means (1935), who presented evidence that different industries had very different patterns of price changes versus quantity changes in the Great Depression. Means showed that simple goods, such as agricultural products, declined heavily in price, while their quantity was almost unchanged. Complex manufactured goods, on the other hand, showed the opposite pattern, with small price changes and consequently huge declines in the quantity of sales. Crude manufactured goods fell somewhere in between. Means's suggestive evidence has led many to speculate on the relationship between output fluctuations and roundabout production; see, for example, Robert J. Gordon (1990).

Means was concerned with the comovement of output and prices during the Depression, but in recent years another stylized fact of business cycles—the procyclicality of productivity—has attracted greater interest. Early work on real business cycles ascribed measured fluctuations in productivity to actual changes in production technology. However, Robert E. Hall (1988) and Charles L. Evans (1992) have shown that

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productivity is correlated with variables that are exogenous with respect to technology. Hall (1988, 1990) explains cyclical productivity as a consequence of imperfect competition and increasing returns. Hall's explanations imply that cyclical changes in sectoral productivity should be a function only of changes in sectoral output. Ricardo J. Caballero and Richard K. Lyons (1990, 1992) document, however, that changes in sectoral productivity are also correlated with aggregate output fluctuations. They interpret this finding as evidence for technological spillovers between sectors. Ben S. Bernanke and Martin L. Parkinson (1991), on the other hand, interpret a similar set of results for sectoral productivity during the Depression as evidence of labor-hoarding.

Here I suggest a new mechanism that explains changes in productivity over the cycle. As in Hall (1988), the explanation relies on imperfect competition. The model differs from Hall's in that it takes account of intermediate goods in production and generates countercyclical markups. Therefore the explanation for cyclical productivity is quite different from Hall's; in particular, even with constant returns in production, the "cost-based Solow residual"—which Hall (1990) shows is the right measure of total factor productivity under imperfect competition—is also procyclical. As a result of these differences, the model predicts that sectoral productivity should in fact be correlated with aggregate activity. I show that this result implies potentially substantial biases in Hall's estimates of the markup and the degree of returns to scale. The explanation I propose can be distinguished empirically from those of Caballero and Lyons (1990, 1992) and Bernanke and Parkinson (1991). If the component of procyclical productivity that appears as an external effect were caused by technological externalities or cyclical factor utilization, this effect should be evident in both gross-output and value-added data. On the other hand, if the explanation I propose here is correct, external effects should appear in value-added data, but not in gross-output data. Performing this test with data from U.S. manufacturing industries supports the model I present here. Other empirical tests also support the predictions of the model.

These are the main results of the paper when it is broadly construed as an example of models of countercyclical markups. The particular model I present, however, generates countercyclical markups by assuming that firms face small costs of changing prices (menu costs) and therefore have rigid prices over some range of shocks. In this setting, modeling firms' use of intermediate goods in production also validates Means's conjecture. For parameter values taken from U.S. manufacturing, the model shows that the roundabout nature of production allows sticky-price models to explain much larger output fluctuations and more severe welfare losses than heretofore thought plausible.[1] One should note, however, that the results of the paper on productivity are independent of whether prices are sticky and depend only on the structure of production combined with imperfect competition and countercyclical markups.

I model the use of intermediate goods in an input–output production structure, so all firms use intermediate inputs in production. If price changes are costly, they are presumably costly for all firms—including those producing intermediate goods—so intermediate goods should also have rigid prices. Intermediate goods, however, act as a multiplier for price stickiness: a little rigidity at the level of an individual firm leads to a large degree of economy-wide price inflexibility.

The reason is straightforward. The representative firm is connected by a complex input–output relationship to many other firms. But each firm cares only about the ratio of its price to its marginal cost of production; an increase in aggregate demand induces a firm to raise its price only to the extent that its profits are squeezed between a fixed output price and rising in-

[1]See the discussions by Laurence Ball and David Romer (1989, 1990) of the early menu-cost models of N. Gregory Mankiw (1985) and George A. Akerlof and Janet L. Yellen (1985).

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put costs. With intermediate goods in production, the increase in firms' costs depends on whether other firms raise prices; so in response to a demand shock, each firm simply "waits by the mailbox" to see if other firms have raised their prices. If other prices go up, then the firm will also be obliged to raise its own price. But if all firms follow this reasonable strategy, no input prices (and hence no output prices) will increase. In the limit, as intermediate goods become the only variable input to production, firms never change prices, and output is determined solely by aggregate demand.[2]

The assumption of sticky intermediate goods prices is supported by the evidence. The most detailed analysis of nominal rigidities studied intermediate goods. George J. Stigler and James K. Kindahl (1970) collected data on actual transaction prices for a large number of such products. Dennis W. Carlton's (1986) analysis of this data set showed that for some substances, particularly steel, paper, chemicals, stone, and glass products, prices can be rigid for long periods of time—in some cases, years.[3] In this paper, I present evidence that prices of intermediate goods are less procyclical than labor costs. This also is consistent with my hypothesis that prices of intermediate goods are relatively rigid.

There is a similarity between the conclusion of this part of the paper and that of Olivier J. Blanchard (1983), but any similarity in the models is more apparent than real. I model price stickiness as state-dependent and the production structure as following an input–output relationship, while Blanchard has time-dependent pricing and in-line production. In Blanchard's model, the degree of price stickiness is a function of the number of stages of processing only because price-setting is assumed to be staggered along the chain of production. If the pricing decision in Blanchard's model were made state-dependent then, since the "first good" is made without intermediate goods, there would be no increase in price rigidity regardless of the number of stages of production. In my state-dependent model, price stickiness depends upon the use of intermediate inputs because the input–output structure of production ensures that all firms use intermediate goods.

Given this difference between the two models, one might ask whether production should in fact be modeled as an input–output process, or as an irreversible chain where goods move in only one direction down the stages of processing. While the "chain of production" seems plausible prima facie, it naturally leads one to ask whether in the real world there are empirically relevant "first goods," the ones produced without any intermediate inputs. Input–output studies certainly do not support the chain-of-production view; even the most detailed input–output tables show surprisingly few zeros.[4] Empirically, the biggest source of any industry's inputs is usually itself: that is, the diagonal entries of input–output matrices are almost always the largest elements

[2]Therefore intermediate goods are a "real rigidity" in the sense in which the term is used by Ball and Romer (1990). However the real rigidities that they provide as examples are all auxiliary assumptions about the behavior of labor or product markets (e.g., efficiency wages and kinked demand curves), whose existence and extent are controversial. By contrast, usage of intermediate goods is a widespread and easily documented feature of the production process in any modern economy.

[3]In fact, Carlton's results imply so much price rigidity that some (including Carlton) have refused to believe that these prices are allocative. Instead, they hold that long-term relationships between buyers and suppliers make the observed spot prices a bad indicator of the true shadow cost of intermediate inputs. If this were true, however, then there should be a strong positive correlation between the observed rigidity of input prices and the length of buyer–seller association. But in fact, Carlton finds a *negative* relationship between price rigidity and the length of association, making it unlikely that the "installment payment" interpretation of price rigidity is correct.

[4]In its discussion of the 1977 input–output table, the BEA (1984 p. 50) notes that the table "shows heavy interdependence among industries. Seventy-six of the [85] industries shown in the table required inputs of at least 40 commodities, and 52 industries required inputs of at least 50 commodities."

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of each column (see Bureau of Economic Analysis, 1984). This seems to lend credence to the view of "roundabout" rather than "in-line" production.

The paper is organized into six sections. Section I presents a simple menu-cost model and uses it to demonstrate that, with intermediate goods, sticky-price models can explain larger output fluctuations. Section II derives reasonable parameter values for the model. Section III shows how the model broadly construed—that is, independent of whether prices are sticky—can generate procyclical productivity movements. Section IV shows how the results derived in the previous section affect Hall's estimates of the markup and the degree of returns to scale. Section V examines empirical evidence from U.S. manufacturing and asks whether the evidence is consistent with the predictions of the model. Section VI contains concluding remarks.

I. The Model Narrowly Construed: Menu Costs

The model is based on Mankiw (1991). There is a continuum of goods, indexed on $[0,1]$. The representative consumer maximizes a utility function that is additively separable in goods, real balances, and leisure:

$$(1) \qquad U = \frac{1}{1-\phi}\int_0^1 Q_{i,\mathrm{F}}^{1-\phi}\,di + \log\left[\frac{M}{P}\right] - L$$

where $Q_{i,\mathrm{F}}$ is the quantity of product i used for final consumption, ϕ is the reciprocal of the elasticity of substitution between different products ($0 < \phi < 1$), M is money demand (assumed equal to money supply by money-market equilibrium), P is the general price level, and L is labor supply. Money is put in the utility function as a shortcut for generating money demand. I have assumed a constant disutility of labor. Note that the assumption of additive separability makes the quantity of any product consumed independent of the prices of all other products.

The price level, P, is defined by

$$P = \left(\int_0^1 P_i^{(\phi-1)/\phi}\,di\right)^{\phi/(\phi-1)}.$$

The price level is, of course, homogeneous of degree 1 in all prices. The consumer maximizes (1) subject to a standard budget constraint; the first-order conditions are derived in the Appendix.

The production side of the economy is composed of a continuum of monopolistic firms, each producing one variety of product. Each firm maximizes profits, given the production function

$$(2) \qquad Q_i = L_i^{\alpha} I_i^{1-\alpha}$$

where

$$I_i \equiv \left(\int_0^1 I_{ki}^{1-\phi}\,dk\right)^{\frac{1}{1-\phi}}$$

and where L_i is the labor input of firm i and I_{ki} is the quantity of the kth intermediate input used by firm i.

I assume that all goods can serve either as final outputs or as inputs for the production of other goods. There is, therefore, no distinction between firms producing manufactured inputs and those producing final goods; all firms produce for both markets. For simplicity, I also assume that firms' elasticity of substitution between manufactured inputs in production is the same as consumers' elasticity of substitution between goods in consumption.[5] Finally, I take the production function to be constant returns to scale and Cobb-Douglas, with the share of nonproduced inputs (here only labor) being α.

[5]This simplification is inessential for any of the results in this paper. Its only purpose is to ensure that firms face a constant-elasticity-of-substitution demand curve for their output.

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Under these conditions, each firm's profit-maximizing nominal price, P_i^*, is

$$(3) \qquad P_i^* = \frac{1}{1-\phi}\left(kW^{\alpha}P^{1-\alpha}\right) \equiv \mu\left(kW^{\alpha}P^{1-\alpha}\right)$$

where μ is the markup, W is the nominal wage, and k is an unimportant constant.[6] The intuition for (3) comes from the fact that the output price is set as a markup on marginal cost; since the production function is Cobb-Douglas, marginal cost is a geometric average of the wage and the overall price of intermediate goods, where the weights are the shares of the inputs in production.

It follows from the equality of the wage and the nominal money supply shown in the Appendix that the optimal relative price for each firm i, p_i^*, is

$$(4) \qquad p_i^* \equiv \frac{P_i^*}{P} = k\mu\left(\frac{M}{P}\right)^{\alpha}.$$

The important point to note about (4) is that the optimal relative price depends upon real balances raised to the power α, rather than to the power 1 as in the simple menu-cost model of Mankiw (1991). Therefore the change in the optimal price following a monetary shock is α times the percentage change in money. Depending on the value of α, this can substantially reduce the loss to a firm that does not adjust its price after a monetary shock, relative to the case where the use of intermediate goods is not modeled. The intuition is straightforward. Aggregate demand is proportional to real money balances, so with fixed prices an increase in money raises output and the demand for labor. Workers are always on their labor supply curves, so the increased use of labor raises the real wage. Since firms set their optimal relative price as a markup on real marginal cost, the increase in the real wage raises the optimal price. But if intermediate goods are used in production, firms' marginal costs rise only in proportion to labor's share, α, since intermediate-goods prices are fixed.

As Mankiw (1985) and Akerlof and Yellen (1985) point out, the loss to a monopolistic firm of not changing its price in response to a money shock is of second order in the change in the optimal price. For price stickiness to be a Nash equilibrium, it must be the case that the loss to each firm of not adjusting prices, *assuming that no other firm adjusts*, is less than the menu cost of changing prices. Each firm calculates the private cost of leaving prices unchanged under the Nash assumption that all other prices will remain at their current levels. The calculations performed in this section show how this private cost of price rigidity varies as a function of the importance of intermediate goods in production.

To a second-order approximation, the change in profit to a firm from not adjusting its price in response to a monetary shock is given by

$$(5) \qquad \pi(p_{\text{new}}^*) - \pi(p_{\text{old}}^*) \cong \tfrac{1}{2}\pi''(p^*)(p_{\text{new}}^* - p_{\text{old}}^*)^2$$

where π'' is the second derivative of the profit function with respect to prices. The difference between p_{new}^* and p_{old}^* is of course proportional to the change in real balances.

The goal is to see how the profit loss (5) varies with respect to α, the share of labor in total cost. As far as the change in the firm's optimal relative price is concerned, this is not difficult. From the discussion above, it is clear that the change is proportional to α. If the loss were based solely on the square of the change in the optimal price, it would diminish strictly in proportion to α^2. However, there are also the changes in the π'' term to be considered. Unfortunately, the expression for the derivative of π'' with respect to α is positive. The two effects work against each other, so it is not possible to sign the derivative unambiguously. Therefore I present nu-

[6] See the Appendix for details.

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TABLE 1—LOSSES FROM A 1-PERCENT MONEY SHOCK (AS PERCENTAGE OF PROFIT)

	α					
	1	0.9	0.7	0.5	0.3	0.1
Loss:	1.000	0.8102	0.4901	0.2500	0.0900	0.0100

Note: Entries give numerical evaluations of equation (6), where $\pi^* - \pi$ is defined in equation (5).

merical evaluations of (5) for different values of α.

The numerical results are reported in Table 1. Table 1 gives the loss to a firm of not adjusting its price in response to a 1-percent money shock as a function of α. The loss is expressed as a fraction of firm profits. To facilitate comparison, the losses are normalized so that the profit loss in the base case ($\alpha = 1$) is 1. Thus each entry in Table 1 reports the quantity

$$(6) \qquad \frac{(\pi^* - \pi)/\pi^*}{[(\pi^* - \pi)/\pi^*]|_{\alpha=1}}$$

where starred variables reflect optimal values and the denominator is always evaluated at $\alpha = 1$.

Firms' losses diminish significantly as α decreases. The results of Table 1 conform to expectation: profit losses fall by approximately a factor of α^2. The loss is not precisely α^2 because the π'' term also changes with α. This second effect is insignificant, however, showing up in the fourth decimal place if at all. (The other important parameter, the elasticity of demand ϕ, just changes the equilibrium levels of output and profits. Variations in ϕ do not affect the loss once it is normalized by the loss in the base case of no intermediate goods, where $\alpha = 1$.)

If one assumes that the firm is indifferent between changing its price and leaving it constant for a 1-percent money shock at $\alpha = 1$, what is the percentage change in money required to leave it similarly indifferent for lower values of α? Since losses are basically proportional to the square of the money shock, the required money shock is given by the square root of the inverse of each entry in Table 1. Looking at the loss for $\alpha = 0.9$, it is interesting to note that even with low use of intermediate goods in production, the allowable money shock jumps by more than 10 percent.

In the next section I suggest reasonable ranges for α and μ, given the share of intermediate inputs in revenue and econometric estimates of the markup. I show that the introduction of intermediate goods with sticky prices into a menu-cost model is quantitatively important. These results follow logically from the assumption of sticky intermediate (and final) goods prices. One might ask, however, whether this assumption is a reasonable description of the world: just how cyclical are the costs of intermediate inputs, particularly relative to the cost of labor? Before answering the question it is necessary to dispel a common misconception. One typically thinks of intermediate inputs as "materials"—raw commodities whose prices are known to be volatile and procyclical.[7] But raw materials and energy are actually only a small fraction of total intermediate input. In a modern economy, by far the largest share of these inputs is devoted to purchases of goods manufactured by other firms. This paper takes the same view as the National Income Accounts: intermediate goods are properly distinguished by use, not by type of good. Once one takes the correct input–output view of intermediate inputs, it is easy to believe that

[7]This confusion results in large part from terminology. In the production-function literature, inputs are classified as KLEM—capital, labor, energy, and materials. That literature takes the correct view of "materials" as all intermediate goods and services, but the word is confusing in this context because it leads one naturally—but incorrectly—to identify "materials" with unprocessed commodities.

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intermediate goods have relatively rigid prices. At the least, the assumption of rigid materials prices is no less reasonable than the assumption of rigid final-goods prices: in many cases, materials *are* final goods.[8] Understanding the correct definition of intermediate inputs helps one understand why Stigler and Kindahl (1970) found that intermediate-goods prices were rigid for long periods of time. In Section V, I present evidence showing that intermediate-goods prices are less procyclical than labor costs. This is further evidence in support of my hypothesis that prices of intermediate goods are relatively rigid.

II. Choosing Reasonable Parameter Values

How can the model be calibrated to judge what are reasonable values of α? In their examination of gross output in U.S. manufacturing, Dale W. Jorgenson et al. (1987) find that the share of intermediate inputs in total manufacturing output is 50 percent or more over the period 1947–1979; so a value of 0.50 seems conservative.

To use the figure cited above, it is necessary to derive a relationship between the revenue share of materials and α. The share of intermediate goods in total revenue is $(1-Q_F/Q)$, where Q_F is final production (or value added), and Q is total (gross) output. Using the equations in the Appendix, one obtains

$$(7) \qquad (1-\alpha)=\mu\left(1-\frac{Q_F}{Q}\right).$$

This equation defines a negative relationship between α and the markup. That, combined with the restriction that α must be between 0 and 1, defines a range of possible values. Thus, the upper bound for α is 0.5 (this corresponds to the case where price equals marginal cost), but with markups the true α will be smaller. The intuition for the appearance of the markup in the expression for the share of intermediate goods comes from the fact that the economy is imperfectly competitive. Since the production function is Cobb-Douglas, the share of intermediate goods in total *cost* is $1-\alpha$. But in an economy with monopolistic competition, the cost share equals the revenue share multiplied by the markup. To calculate the cost share from the observable revenue share, one must take a position on the size of the markup.

The right concept of the markup for this paper is the ratio of the price of gross output (not real value added) to its marginal cost of production. As Ian Domowitz et al. (1988) note, if materials are used in production, then the markup estimated from value added exceeds the true markup because it divides profits by a smaller denominator (value added rather than gross output). I argue below that estimating the markup using gross-output data also avoids other biases of value-added data. Using gross-output data, Domowitz et al. estimate an average markup of 1.6 for the industries in their sample. This makes it possible to pin down the relevant value of α.

Table 2 gives a range of values for α, with corresponding implied values of μ. Each column is computed for a different value of α. The first line of the table reports the markups corresponding to these values of α, given the observed revenue share of materials of 0.5. The next line of the table gives the profit loss from price stickiness for a 1-percent money shock, normalized as before so that it is a fraction of the profit loss for $\alpha=1$.

Given an estimated markup of 1.6, the relevant columns of Table 2 are the last two on the right, corresponding to $\mu=1.5$ and $\mu=1.7$. These show profit losses to firms declining by a factor of 25–100. What are the consequences for business cycles? The

[8] One commonly hears the claim that Blanchard (1983) and Kevin M. Murphy et al. (1989) have shown that intermediate input prices are more procyclical than final-goods prices. Their conclusions result from incorrectly equating intermediate goods with unprocessed goods.

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TABLE 2—LOSSES FROM A 1-PERCENT MONEY SHOCK FOR CALIBRATED PARAMETER VALUES

	α				
Variable	0.45	0.4	0.3	0.2	0.1
Implied markup, μ[a]	1.042	1.137	1.327	1.516	1.706
Loss from fixed prices for a 1-percent money shock[b]	0.202	0.160	0.090	0.040	0.010

[a]Derived from equation (7), assuming $1 - Q_F/Q = 0.5$.
[b]Numerical evaluations of equation (6), where $\pi^* - \pi$ is defined in equation (5).

size of the maximum shock to money for which nonadjustment is a Nash equilibrium jumps five- to tenfold, implying larger output fluctuations from sticky prices. Thus, once one recognizes the role of intermediate goods in production, menu-cost models can explain significantly larger business cycles.

The welfare consequences are also immediate. Ball and Romer (1989, 1990) summarize welfare by examining the ratio, R, of the social cost of output fluctuations to the private cost. Since, in this model, the social cost of fluctuations comes solely from the disutility of variance in consumption, it is unaffected by introducing intermediate goods. But as I have shown, for plausible parameter values the private cost of business cycles falls considerably. Therefore, R increases by a factor of 25–100, so menu-cost models can also explain inefficient business cycles.

Are the calibrated parameter values reasonable? One might argue that if the markup is as high as 1.6, the share of profits in output becomes implausibly large. The most natural way to reconcile high markups with low observed profits is to suppose that there are large fixed costs of production. In this view, output in excess of variable cost is largely consumed by fixed costs.[9] The production function of equation (2) can easily be amended to allow for fixed costs, without any change in the preceding analysis. With fixed costs, however, the interpretation of α changes: α is no longer the share of labor in total cost, but the share of *variable* labor in total *variable* cost. This is important to keep in mind, because I have argued that α is in the range of 0.1–0.2. But the share of nonproduced inputs (capital and labor) in total cost is roughly 0.5. With overhead labor and capital, however, these different figures are not contradictory: it is perfectly possible for the share of variable labor and capital to be small, while their total share (inclusive of fixed costs) is large.

[9]This is the view of Hall (1986), and is true in Chamberlinian monopolistically competitive equilibrium.

III. The Model Broadly Construed: Countercyclical Markups

This section shows that a business-cycle model with constant returns in production, no technological externalities, and no technology shocks can account for one of the major stylized facts of business cycles, procyclical labor and total factor productivity. As noted in the Introduction, the result of procyclical productivity and its consequences, which are explored in the following sections, would obtain in any model with intermediate goods and countercyclical markups. Price rigidity is one way of generating countercyclical markups, but other explanations—such as the customer-market model of Edmund S. Phelps and Sidney G. Winter (1970), or the supergame-theoretic model of Julio J. Rotemberg and Garth Saloner (1986)—would serve equally well.

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Thus, the results of this section and the following ones apply to a much broader class of models than the one considered in Section I and constitute an interesting mechanism for the transmission of shocks in purely real models of countercyclical markups, such as Rotemberg and Michael Woodford (1991).

The result of procyclical productivity is driven by three properties of the model. First, since firms are imperfectly competitive (prices are above marginal costs), the equilibrium is inefficient. Second, the inefficiency is lower at higher levels of output; the model has the feature that markups are countercyclical. Third, intermediate goods are used in production. Blanchard and Nobuhiro Kiyotaki (1987) present a model that has the first and second features, but not the third. In their model, the presence of markup pricing distorts the labor–leisure choice but not producers' decisions about the mix of inputs to employ: since Blanchard and Kiyotaki assume that labor is the only input to production, there is no "input mix" to distort. Thus, even though higher levels of output raise welfare in their model, this increase in welfare does not raise productivity. But when manufactured inputs are used in production, markup pricing also makes firms' production decisions socially suboptimal. In particular, at the initial equilibrium, firms use too much of the primary input (labor) and too few manufactured inputs. In this model, the decrease in markups that accompanies output movements causes the ratio of input prices to be closer to the marginal rate of transformation between goods and labor. Thus, firms making their input choices make decisions about the quantity of labor to employ versus the quantity of manufactured inputs to use that are closer to being socially optimal. This increase in social efficiency not only increases welfare, it also raises productivity. So in this model there are endogenous variations in productive efficiency, caused by the fact that the economy is moving closer to the boundary of its production-possibilities frontier during a boom and farther away during a recession.

So far the discussion of the paper has used gross output as the concept of production. However, gross output is the total output of a firm, including the output used by other firms as intermediate goods. But the efficiency of an economy is judged by its ability to produce final goods from a given quantity of nonproduced inputs (here, only labor). Therefore, the correct statistic to examine is value added relative to labor, rather than gross output relative to labor.

To examine the issue of cyclical productivity, I first derive the economy-wide (and sectoral) value-added production function. Since labor is the only input to production and there are constant returns to scale, one can represent the net output or value-added production function as

$$Q_F = AL \tag{8}$$

where A is this economy's "total factor productivity."

The objective is to examine the change in productivity as a function of the change in value added, assuming that menu costs are large enough to prevent prices from changing in response to the money shock that causes the output expansion. Alternatively, one can view the output movement as an expansion of real aggregate demand in a flexible-price model where, for any of the reasons given in the models cited above, a 1-percent increase in final output is accompanied by an $\alpha\phi$-percent reduction in the markup.

The percentage change in A is derived from this experiment. Taking the appropriate derivatives and evaluating the resulting expression at the real wage that prevails at the initial equilibrium yields

$$\frac{dA}{A} = \left(\frac{\phi(1-\alpha)(\mu-1)}{\alpha+(\mu-1)} \right) \frac{dQ_F}{Q_F}. \tag{9}$$

This expression is positive and rises monotonically as α falls. It shows why imperfect competition is necessary for the result: if

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TABLE 3—CYCLICALITY OF PRODUCTIVITY

	α				
Variable	0.45	0.4	0.3	0.2	0.1
Implied markup, μ[a]	1.042	1.137	1.327	1.516	1.706
Percentage change in productivity for a 1-percent output change[b]	0.002	0.018	0.090	0.196	0.326

[a]Derived from equation (7), assuming $1 - Q_F/Q = 0.5$.
[b]Numerical evaluations of equation (9).

$\mu = 1$, so that there are no distortions in production, productivity is not procyclical. Of course, if $\alpha = 1$ there are no intermediate goods, and productivity is again acyclical: this is the special case of Blanchard and Kiyotaki (1987). Note that the result is *not* being driven by Hall's (1988) argument that the Solow residual is procyclical if there are markups. In my model economy, dA/A is the correct measure of the change in productive efficiency; in Hall's (1990) terminology it is the "cost-based Solow residual." Hall (1990) claims that this cost-based residual should be invariant even with markup pricing, unless firms have increasing returns to scale. Here I present a counterexample: in this model all firms produce with constant returns to scale, but the cost-based residual is procyclical. The intuition for this result, and its consequences for Hall's tests, are discussed below.

Values for the coefficient in (9) are reported in Table 3 for settings of α and μ that satisfy calibration. The table shows that productivity changes are higher if α is lower. Thus, as one would expect, changes in total factor productivity are larger as intermediate goods become more important in production. For the United States over the period 1962–1984 a 1-percent growth in output resulting from a demand shock is associated with a 0.59-percent growth in total factor productivity (Rotemberg and Lawrence Summers, 1990 table II). For markups around 1.6, the model as calibrated would predict a growth in total factor productivity of between 0.20 percent and 0.33 percent in response to a 1-percent increase in output. Thus, for some empirically reasonable parameter values, the model is capable of accounting for a significant proportion of demand-induced changes in total factor productivity.

A striking feature of this model is that it predicts a positive correlation between sectoral productivity and aggregate activity. The increase in productivity comes from the fact that each industry is making more efficient choices about the mix of factors to employ in production.[10] Given the assumption that each industry uses as intermediate goods mostly the outputs of other industries, the increase in efficiency depends on an across-the-board reduction in markups. A uniform reduction in the markup for all varieties of intermediate goods will be correlated with changes in aggregate activity, but not with industry-specific changes in output. Caballero and Lyons (1990, 1992) present empirical evidence that sectoral productivity is in fact correlated with demand-driven changes in aggregate output in both Euro-

[10]Here is another point where taking the correct, broad view of intermediate inputs provides better intuition. By using a Cobb-Douglas production function, I have assumed that the elasticity of substitution between labor and intermediate goods is 1. Rotemberg and Woodford (1992) use econometric estimates and an imperfectly competitive general-equilibrium model to calibrate this elasticity equal to 1.2—*greater* than the Cobb-Douglas case. These relatively large elasticities of substitution are not surprising when one recalls that intermediate inputs include inputs of services, which are an increasingly large share of total inputs. The elasticity of substitution between using, for example, an in-house computer technician and an outside repairman is surely very high.

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pean and U.S. manufacturing.[11] The model provides an economic explanation of their finding.

This explanation of the Caballero-Lyons stylized fact does not assume that there are true technological spillovers operating at business-cycle frequencies. Not only is it difficult to model such externalities, it is difficult even to tell an intuitive story for what form they might take. The advantage of this model is that it does not rely on high-frequency shifts in the production function, but rather on cyclical changes in the relative price of inputs to production.

Consequently, the model delivers a sharp prediction about how one can distinguish between these two explanations for the Caballero-Lyons findings. It predicts that such an effect should be found when a production function is estimated with value-added data, but not with gross-output data. Estimating the gross-output production function amounts to estimating equation (2). By assumption, there are no technological externalities in the production of gross output, so correct estimation of (2) will reveal none. But each productive unit becomes more efficient at creating value added, because the markup is smaller. This increase in efficiency is correlated with increases in aggregate output. This distinction between the two explanations is testable; it is examined in Section V.[12]

IV. Implications for Hall's Tests

In a series of papers, Hall (1986, 1988, 1990) has proposed various ingenious methods of using time-series data on productivity to determine the markup of price over marginal cost and the degree of returns to scale. One implication of the previous discussion is that, if intermediate goods are used in production and markups are countercyclical, many of Hall's estimates are likely to be biased upward.

Intuitively, Hall's tests assume that true productivity (as measured by the cost-based Solow residual) can be procyclical for only two reasons: technology shocks and increasing returns to scale. I have shown in the previous section that productivity can be procyclical for a third reason: even with constant returns to scale, productivity increases if a demand-driven expansion causes markups to fall. In econometric terms, there is a second error term in Hall's regressions that is correlated with his right-hand-side variables and with his aggregate demand instruments. This second error term is the source of the bias.

It is easy to show that in this model economy characterized by imperfect competition, constant returns, and markup pricing —just what Hall (1988) assumes—Hall's methodology leads to a systematic upward bias in the estimate of the markup. Hall estimates a relationship like

$$(10) \qquad (\Delta \ln Q_F) = \mu^*(\sigma \Delta \ln L)$$

where σ is the share of labor in value added and μ^* is claimed to be an unbiased estimate of the markup.[13]

In the model developed here, however, $\Delta \ln Q_F = \Delta \ln A + \Delta \ln L$. If $\Delta \ln A$ did not comove with output or labor input—or were uncorrelated with demand-driven changes in labor input, which would happen if $\Delta \ln A$

[11] Their use of aggregate demand instruments ensures that the results are not being driven by common or sectoral productivity shocks, as in the model of John B. Long and Charles I. Plosser (1983).

[12] This predicted difference will also hold if the Caballero-Lyons results in fact stem from cyclical factor utilization. A change in utilization is a shift of the gross-output production function and will show up as such in gross-output data as well as in value-added data.

[13] Hall's (1988) procedure treats both capital and labor as inputs into the production of value added. The model aggregates both factors into a single nonproduced input, termed "labor" in honor of its primary component. I do not explicitly model Hall's (1988) instrumental-variables (IV) estimation, because by hypothesis there are no true technology shocks in the model. Hall's IV procedure was designed to purge the explanatory variable of its correlation with technology shocks; using aggregate demand instruments would not solve the problem I identify, since in my model productivity changes with aggregate demand.

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TABLE 4—PREDICTED BIASES IN ESTIMATES OF MARKUPS AND RETURNS TO SCALE

	α				
Variable	0.45	0.4	0.3	0.2	0.1
Implied markup, μ[a]	1.042	1.137	1.327	1.516	1.706
Percentage bias in conventionally estimated μ^{VA} and γ[b]	0.165	1.88	9.87	24.39	48.37

[a]Derived from equation (7), assuming $1 - Q_F/Q = 0.5$.
[b]Numerical evaluations of equation (13).

were a pure technology shock—the expectation of the estimated markup would be

$$(11)\quad E(\mu^* \mid \Delta \ln A = 0) = \frac{\mathrm{Cov}(\Delta \ln Q_F, \sigma \Delta \ln L)}{\mathrm{Var}(\sigma \Delta \ln L)} = \frac{1}{\sigma} = \frac{P}{W/A} = \frac{\mu - (1-\alpha)}{1-(1-\alpha)} \equiv \mu^{VA}.$$

Even without considering the problems posed by the correlation of $\Delta \ln A$ with changes in labor input, one sees that the expectation of μ^* is not equal to the true μ. This is because Hall (1988) estimates the markup on real value added (μ^{VA}) rather than the markup on output.[14] But given an unbiased estimate of μ^{VA}, it is relatively easy to back out the true μ, since the relation between the two depends only on the observable share of intermediate goods in revenue. Thus, if changes in true sectoral productivity ($\Delta \ln A$) are uncorrelated with aggregate demand instruments, Hall's methodology at least gives an unbiased estimate of μ^{VA}, from which one can calculate the true markup, μ.

As shown in (9), however, changes in the level of productivity covary systematically and positively with changes in Q_F and L. Therefore, if one takes into account the fact that $\Delta \ln A$ changes predictably in response to demand shocks, Hall's (1988) methodology creates an upward bias even in the estimate of μ^{VA} in this economy. The size of the bias can be computed from the equation for the true expectation of μ^*:

$$(12)\quad E(\mu^*) = \frac{\mathrm{Cov}(\Delta \ln L + \Delta \ln A, \sigma \Delta \ln L)}{\mathrm{Var}(\sigma \Delta \ln L)} = \mu^{VA}\left(1 + \frac{\mathrm{Cov}(\Delta \ln A, \Delta \ln L)}{\mathrm{Var}(\Delta \ln L)}\right).$$

Evidently the bias is a positive one (since productivity comoves positively with labor input). Calculations of the size of this bias for the calibrated parameter values are given in Table 4. Over the relevant range of parameters (true gross-output markups around 1.6), the upward bias is between 25 percent and 50 percent.

Hall's 1990 paper tested the invariance of the cost-based Solow residual at both the economy-wide and two-digit SIC levels. In this model, Hall's estimating equation would be

$$(13)\quad \Delta \ln Q_F = \gamma(\sigma_c \Delta \ln L)$$

where σ_c is the share of labor in the total cost of producing value added (here fixed at 1) and γ is Hall's estimate of the degree of

[14]Hall is clearly aware of this aspect of the value-added/gross-output distinction; see Hall (1986).

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returns to scale. For the reasons given above and in the previous section, this methodology would wrongly conclude that firms have increasing returns to scale ($\gamma > 1$), when the fact the true γ is identically equal to 1. The percentage biases would be those given in Table 4. An easy way to test the hypothesis that Hall's (1990) estimate of γ is biased upward is to estimate (13) using gross-output data (which should not be subject to this bias) and to contrast the results with Hall's value-added estimates. Below I present evidence indicating that Hall's (1990) estimates are in fact subject to precisely this bias. The gross-output estimates imply returns to scale that are about constant (or slightly decreasing)—a sharp contrast with Hall's (1990) finding of strongly increasing returns, and evidence in favor of this model.

The reason for the bias is the confusion of external effects with internal increasing returns to scale. In the model considered here, all firms produce with constant returns, but an aggregate demand shock increases both output and productivity. Thus, an increase in the output of every firm is correlated with an increase in the economy-wide efficiency of production—an external effect that Hall's procedure mistakes for increasing returns to scale at the firm level.

V. Some Empirical Evidence

The model has two types of empirical implications. First, it predicts cyclical movements of some ratios that are not often studied in business-cycle theory. For example, it implies that the prices of intermediate goods should be countercyclical relative to the price of labor. Also, the quantities of intermediate goods used should be procyclical, again relative to labor input.

More direct tests of the model examine its predictions regarding the substitution of materials for labor and the behavior of total factor productivity.[15] First, the model predicts that there should be a positive correlation between changes in the materials–output ratio and changes in the ratio of wages to the price of intermediate inputs (since the increased use of materials results from a change in the relative price of inputs). Second, if the change in the relative price has the effect claimed in the paper, then there should also be a positive correlation between procyclical total factor productivity and procyclical usage of materials relative to labor. The most novel prediction of the paper is the third effect: as discussed above, estimates of external effects using value-added data should be significant, but similar estimates using gross-output data should be insignificant.

In this section, I check these predictions against U.S. time series data. The data generally support the predictions of the model.

Dale Jorgenson kindly supplied the data I use, which are thoroughly described in Jorgenson et al. (1987). A major improvement of this data set relative to standard National Income and Product Accounts data is that all the inputs are quality-adjusted. The labor-input series, for example, is a quality-weighted index of hours worked by different categories of workers, rather than the usual measure of the sum of hours worked by all workers which implicitly assigns equal weight to all workers.

The data set employs gross production as the relevant concept of output and therefore reports quantities used of capital, labor, energy, and materials. The Jorgenson data set is especially suitable because the model makes predictions about materials usage over the business cycle. I have modified the materials series to reflect usage per year rather than purchases. Although work in process and intermediate-goods inventories are strongly procyclical, as Valerie A. Ramey (1989) documents, the results are not sensitive to this modification (inventories are small relative to total intermediate input use).

The data used for the tests are a panel of annual observations on 21 manufacturing industries in the United States from 1959 to 1984. The definitions of the industries are standard two-digit SIC classifications with

[15] I thank an anonymous referee for suggesting the first two tests.

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TABLE 5—EMPIRICAL REGULARITIES

Dependent variable	Explanatory variables					
	ΔY_i	ΔY	PNP_i	OVT_i	AGH_i	$\Delta P_{\ell,i} - \Delta P_{m,i}$
$\Delta P_{m,i} - \Delta P_{\ell,i}$	−0.20					
	(0.02)					
$\Delta M_i - \Delta L_i$	0.56					
	(0.08)					
	0.41		0.004	0.006	0.026	
	(0.07)		(0.002)	(0.004)	(0.008)	
$\Delta M_i - \Delta Y_i$		0.18				
		(0.02)				
						0.12
						(0.01)

Notes: Standard errors are given in parentheses. The sample period is 1959–1984. M_i and L_i are industry materials and labor input; $P_{m,i}$ and $P_{\ell,i}$ are the associated prices. Y_i and Y are industry and manufacturing gross output. PNP_i, OVT_i, and AGH_i are, respectively, the ratio of production to nonproduction workers, the average number of overtime hours worked, and the number of hours worked by an average worker in each industry. All variables are in logs.

the exception that the Jorgenson data set separates motor vehicles (SIC 371) from other transportation equipment (SIC 372-79). Thus, there are 21 industries, rather than the usual 20.

All of the regressions involve testing for cyclical effects. To avoid the possibility that the cyclicality of both output and, say, the price of materials relative to labor is driven by technology shocks that make materials usage more attractive in a boom, I typically instrument the right-hand side variables. The instruments are those suggested by Ramey (1989): the change in the world price of oil, changes in military expenditures, and the political party of the president.

A. *Cyclical Regularities*

As noted above, one of the predictions of the model is that materials prices are countercyclical, relative to the prices of substitutes such as labor and capital.[16] This prediction is tested using the Jorgenson data. I estimate the following equation (all variables are in logs):

$$(14) \qquad (\Delta P_{m,it} - \Delta P_{\ell,it}) = \text{constant}_i + \beta_1 \Delta Y_{it}$$

where $P_{m,i}$ and $P_{\ell,i}$ are the prices of intermediate goods and labor inputs to a given industry i and Y_i is sectoral output.

Table 5 reports the estimate of β_1 (with the elasticity constrained to be equal across industries). The elasticity is negative and significant, as the model predicts.[17]

This result, that one measure of the real wage is significantly procyclical, may seem at odds with the conventional wisdom that the real wage is acyclical or only slightly procyclical. The cost of labor input is more

[16]The usual claim that the observed wage is less procyclical than the shadow wage strengthens the results of this section. If the true marginal wage is more procyclical than it seems—perhaps because of overtime payments, as stressed by Mark Bils (1987)—then the relative price of materials is even more countercyclical than I find.

[17]"Labor," as used in this model, comprises both capital and labor inputs; "materials" may or may not include energy. In the empirical work I have used the standard definition of labor and a concept of materials that excludes energy (which anyway is only about 5 percent of a typical sector's materials input). Excluding energy shows that the results are not being driven by the oil price shocks of the 1970's. However, the findings are robust to using all combinations of these different concepts; the results are often strengthened by using broader definitions of labor and materials.

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procyclical in the Jorgenson data set because labor quality is significantly countercyclical.[18] A slightly higher real wage paid to lower-quality workers implies that the cost of an efficiency unit of labor is much higher in booms. Thus, in an expansion, labor becomes more expensive relative to intermediate inputs, which must lead producers to economize on labor and use intermediate goods more intensively.

The next prediction I check is the claim that materials usage is procyclical relative to labor. The basic equation I estimate is:

$$(15) \qquad (\Delta M_{it} - \Delta L_{it}) = \text{constant}_i + \beta_2 \Delta Y_{it}$$

where M_i and L_i are (the logs of) intermediate goods and labor inputs. Table 5 gives the estimate of β_2 for U.S. manufacturing. The elasticity is positive and significant.

One might believe, however, that the result of procyclical intermediate-goods usage is being driven by labor-hoarding. Estimates of production functions and the degree of returns to scale are often thought to be subject to cyclical measurement error. In this view, the apparent acyclicality of labor hours may stem from unmeasured procyclical work effort. If so, true labor input will be procyclical even if measured hours are not. If this effect is not taken into account, one might wrongly conclude that the ratio of intermediate goods to labor is procyclical.

One way to control for labor-hoarding is to include right-hand-side variables that are plausible proxies for cyclical labor utilization. One such variable, average hours worked per employee (AGH), has been proposed by Thomas A. Abbott et al. (1989). Following Caballero and Lyons (1992), I also use two other variables to control for labor-hoarding: the ratio of production to nonproduction workers (PNP), and the average number of overtime hours worked (OVT).

The results are found in Table 5. They confirm the hypothesis of labor-hoarding: the variables that control for changes in effective labor input always have the correct sign and are usually significant. As expected, taking labor-hoarding into account reduces β_2. But even accounting for labor-hoarding, intermediate input usage remains strongly procyclical relative to labor.

Another way to see whether labor-hoarding is responsible for the results is to examine the cyclicality of the ratio of intermediate inputs to industry output. In fact, this test is biased against finding procyclicality. Changes in both industry and aggregate output are likely to be driven by common productivity shocks and oil price shocks, imparting a negative bias to the results. For this reason, I do not instrument the explanatory variable, changes in aggregate output. Almost all of the explanatory power of the instruments used previously comes from oil prices. But using oil prices as an instrument would only exacerbate the bias, by isolating those changes in aggregate output that are most strongly correlated with the error term.

The results are also reported in Table 5. The ratio of intermediate inputs to output is procyclical and statistically significant. The numerical magnitudes are smaller, but this is not surprising, given the negative bias in the results noted above. These results should be thought of as a lower bound on the procyclicality of intermediate-goods usage; it is apparent that even the lower bound is positive. The existence of labor-hoarding does not alter this basic finding.

B. *Specific Predictions*

In this section I test the more specific predictions of the model. First, I test whether changes in the intermediate-goods–output ratio are consistently related to changes in the relative price of these inputs. To test the prediction, I run the

[18] Finn E. Kydland and Edward C. Prescott (1988) and Gary Solon et al. (1994) show that this composition effect is important for analyzing real-wage cyclicality.

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regression

$$(16)\quad (\Delta M_{it} - \Delta Y_{it}) = \text{constant}_i + \beta_3(\Delta P_{\ell,it} - \Delta P_{\text{m},it}).$$

The result is found in the last line of Table 5: β_3 is positive and significant. Note that if the production function is in fact Cobb-Douglas, the estimated coefficient in this regression should be α. The coefficient is 0.12, which is in line with the calibrated value of α: based on evidence regarding the size of the markup, α was predicted to be between 0.2 and 0.1.

Next I test the prediction that changes in the input mix are responsible for changes in total factor productivity. Here, however, there is the problem with labor-hoarding discussed above. If there is a significant degree of unmeasured factor utilization that applies to labor but not to intermediate goods, the measured procyclicality of total factor productivity and of the intermediate-goods–labor ratio may both be driven by cyclical measurement error. There is a way to distinguish these two hypotheses, however. To the extent that cyclical measurement error is driving the finding of procyclical productivity, this effect should be apparent in both gross-output and value-added data. However, as argued above, the procyclicality resulting from countercyclical markups should be found only in value added. This suggests that we should regress two different measures of the Solow residual on changes in the materials–labor ratio: one measure calculated from gross output and the other from value added. If the value-added estimate is significantly larger, this will imply that the mechanism identified in the paper is at work. Therefore, I estimate

$$(17)\quad \Delta\text{TFP}_{it} = \text{constant}_i + \beta_4(\Delta M_{it} - \Delta L_{it}).$$

In calculating the growth rate of total factor productivity, I use cost shares[19] rather than revenue shares to avoid the problem pointed out by Hall (1988): productivity calculated using revenue shares appears spuriously procyclical if firms price their product above marginal cost. I do not allow for increasing returns in production, which Hall (1990) argues is responsible for the failure of invariance in the cost-based Solow residual. Recent empirical work by Martin N. Baily et al. (1992) applies Hall's procedure to plant-level gross-output data from the Longitudinal Research Database and finds essentially constant returns to scale. In the next series of tests, I come to the same conclusion using industry-level gross-output data.

The results of the test using total factor productivity are reported in Table 6. Note first that there is evidence of significant procyclicality of the gross-output residual ($\Delta\text{TFP}^{\text{GO}}$) in response to changes in the materials–labor ratio. Therefore, as indicated above, part of the movement in this ratio most likely reflects changes in unmeasured labor utilization. However, it is also clear that the value-added estimate significantly exceeds (by almost a factor of 3) the gross-output estimate. Therefore, although it appears that some of the correlation between total factor productivity and changes in the materials–labor ratio reflects labor-hoarding, the data support the contention that some other mechanism like the one proposed here is also at work.

Finally, I test the prediction that, if the mechanism proposed by the model is responsible for the finding that procyclical productivity is an external effect, we should be able to detect the effect in value-added data but not in gross-output data.[20] The

[19]The cost shares are calculated as in Hall (1990). However, I use capital-specific depreciation rates and tax parameters (the investment tax credit and the value of depreciation allowances) that vary by industry.

[20]A different explanation is proposed in Basu and John G. Fernald (1994a,b) for why value-added data should give incorrect estimates of returns to scale and external effects. That explanation basically rests on the correct claim that with imperfect competition, value added should be calculated using the cost share of materials rather than the revenue share. (For the definition of value added, see Kenneth J. Arrow [1974].) To meet this objection, I used cost shares to construct the value-added data that I use in equation (18). As the results show, even with this correction there are external effects in value added and none in gross output, as the model predicts.

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TABLE 6—VALUE-ADDED AND GROSS-OUTPUT RESULTS

Dependent variable	Explanatory variables				
	$\Delta M_i - \Delta L_i$	ΔX_i^{VA}	ΔX^{VA}	ΔX_i^{GO}	ΔX^{GO}
ΔTFP_i^{VA}	0.33				
	(0.02)				
ΔTFP_i^{GO}	0.12				
	(0.01)				
ΔY_i^{VA}		0.63	0.79		
		(0.04)	(0.08)		
ΔY_i^{GO}				0.96	0.01
				(0.01)	(0.02)

Notes: Standard errors are given in parentheses. The sample period is 1959–1984. ΔY_i^{VA} and ΔY_i^{GO} are growth rates of industry value added and gross output. ΔX_i^{VA} is the sum of the growth rates of industry capital and labor inputs, each weighted by its cost share in the production of value added; ΔX^{VA} is the analogue for aggregate manufacturing. ΔX_i^{GO} is the sum of the growth rates of industry capital, labor, energy, and materials inputs, each weighted by its cost share in the production of gross output; ΔX^{GO} is the analogue for aggregate manufacturing. ΔTFP_i^{VA} is the growth rate of industry total factor productivity calculated from value added and capital and labor inputs: $\Delta TFP_i^{VA} = \Delta Y_i^{VA} - \Delta X_i^{VA}$. ΔTFP_i^{GO} is the growth rate of industry total factor productivity calculated from gross output and capital, labor, energy, and materials inputs: $\Delta TFP_i^{GO} = \Delta Y_i^{GO} - \Delta X_i^{GO}$. ΔM_i and ΔL_i are growth rates of industry materials and labor input.

empirical procedure follows Caballero and Lyons (1989). I estimate the equation:

$$(18) \quad \Delta Y_{it} = \text{constant}_i + \gamma \Delta X_{it} + \kappa \Delta X_t$$

where ΔX_i is the cost share-weighted sum of sectoral input growths and ΔX is growth of aggregate (manufacturing) inputs, similarly cost-weighted. ΔY_i^{VA} is the growth of value added; ΔY_i^{GO} is the growth of gross output. In the value-added regressions, the inputs are capital and labor; in the gross-output regressions the inputs are capital, labor, energy, and materials. Value added is constructed using the cost share of materials. Finally, γ is the degree of internal returns to scale, and κ captures external effects from aggregate activity. (See Caballero and Lyons [1989] for a fuller description of the procedure.)

There is a significant difference between the value-added and gross-output estimates. Table 6 shows that in the gross-output regressions the point estimate of κ is 0.01—close to zero, and not significantly different from zero. On the other hand, the value-added estimate is 0.79, with a t statistic that exceeds 9. This finding is significant for interpreting the Caballero-Lyons stylized fact. The difference between the two sets of results indicates that their finding of a large positive κ is not evidence for a true technological externality, but rather an indication that a more subtle effect, perhaps having to do with cyclical changes in markups, is at work. Note that the estimate of γ, the degree of internal returns to scale, is essentially 1 in the gross-output data, implying that there are constant returns to scale. This finding contrasts with the results of Hall (1990), who finds significantly increasing returns using value-added data. As the previous section shows, the model presented here can explain this divergence.

This simple model therefore has a number of predictions about cyclical patterns of input use, input prices, and the behavior of productivity over the business cycle. Data from U.S. manufacturing industries generally confirm these predictions.

VI. Conclusion

A wide variety of evidence indicates that modern economies are characterized by imperfectly competitive behavior, and many

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business-cycle models of imperfect competition imply countercyclical markups. Intermediate goods are widely used in production. This paper has explored the implications of these two sets of stylized facts and has found that in conjunction they lead to a number of strong results. In purely real models they imply that productivity, even correctly measured, is procyclical in response to demand shocks, even with constant returns to scale in production and no technological externalities. In models where countercyclical markups are a consequence of output price rigidity, modeling the use of intermediate goods in production implies that business cycles are both larger and more costly.

The model makes a number of predictions about cyclical productivity that accord with the facts. Among other things, it predicts that sectoral productivity should appear in the data as an external effect: the productivity of one sector should be correlated with aggregate rather than sectoral activity. The model also implies that Hall's (1986, 1988, 1990) estimates of markups and returns to scale are biased upward. Other authors have interpreted the finding of external effects in productivity as evidence for technological spillovers or for labor-hoarding. I show that there is a sharp empirical test that can discriminate among these various hypotheses. If the type of model presented here is responsible for the finding of external effects, these effects should be present in value-added data but not in gross-output data. If the spillover or labor-hoarding hypotheses are at work, on the other hand, then they should be present in both gross output and value added. It turns out that the spillovers are found only in value added, which confirms the predictions of the model. The paper predicts that the biases in Hall's work should also be a function of his use of value-added data. Using Hall's procedure to estimate returns to scale from gross-output data shows that there are constant returns, not the strongly increasing returns that Hall finds. This finding also supports a model of the kind presented here, with countercyclical markups and intermediate goods in production.

It is important to stress that, although price rigidity is not necessary to derive the results on cyclical productivity, these results do follow naturally from a sticky-price model. Thus, a setting in which the menu-cost assumption easily explains large welfare losses (a model with imperfect competition and heavy usage of intermediate goods) also enables these models to explain many of the stylized facts on cyclical productivity. This paper, then, provides a link between the purely real and purely nominal literatures within the new-Keynesian economics. This link should be a subject of future research.

Appendix: Derivations

Maximizing (1) subject to a standard budget constraint gives the consumer's first-order conditions:

$$W = M \tag{A1}$$

and

$$Q_{i,\mathrm{F}} = W^{1/\phi} P_i^{-1/\phi}. \tag{A2}$$

Minimizing costs subject to the production function (2) gives the input demands for each firm:

$$I_{ki} = \left(\frac{P_k}{P}\right)^{-1/\phi} \left(\frac{(1-\alpha)W}{\alpha P}\right) L_i \tag{A3}$$

and

$$L_i = \left[\left(\frac{\alpha}{1-\alpha}\right)\frac{P}{W}\right]^{1-\alpha} Q_i. \tag{A4}$$

The cost function is then given by substituting (A3) and (A4) into the expression for the cost of production:

$$C_i = kWQ_i\left(\frac{P}{W}\right)^{1-\alpha} \tag{A5}$$

where

$$k = \left(\frac{\alpha}{1-\alpha}\right)^{1-\alpha} + \left(\frac{\alpha}{1-\alpha}\right)^{-\alpha}.$$

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The total output of each firm, Q_i, is given by the sum of demands for its output as final goods [equation (A2)] and as intermediate inputs [the integral of (A3) over i]. This gives

$$(A6) \quad Q_i = \left[\left(\frac{\alpha}{1-\alpha}\right)^{-\alpha}\left(\frac{W}{P}\right)^{\alpha} P^{1/\phi} Q + W^{1/\phi}\right] P_i^{-1/\phi}$$

where $Q = \int_0^1 Q_i \, di$ is aggregate (gross) output. One can solve for the equilibrium aggregates Q_F and Q from (A2) and (A6), using the fact that all firms' quantities and prices are equal in this symmetric equilibrium.

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